Lab 2 Activity Sheet

BEE 4750/5750, Fall 2026

Name: _____

NetID: _____

This sheet covers the written parts of Lab 2 (Linear Programming with JuMP). Do the coding in `lab02.ipynb` or `lab02.jl`.

A utility owns three power plants side by side on a river. It sells all they generate at \$120/MWh this hour and chooses each plant's output to make the most profit. The coal and combined-cycle plants are cooled with river water; the combustion turbines are air-cooled. The three plants share three limits this hour: at most 1,000 MWth of heat discharged to the river (MWth is megawatts of heat, as distinct from MW of electricity), at most 650 lb of NOx, and at most 1,000 MW on the transmission line they share.

Plant	Capacity (MW)	Variable cost (\$/MWh)	Heat to river (MWth per MW)	NOx (lb/MWh)
Coal steam	600	30	1.5	1
Combined cycle	400	56	0.64	0.1
Combustion turbines	300	87	0	0.5

Problem 1: Formulate the LP

Let y_1, y_2 and y_3 be the output of the coal plant, the combined-cycle plant and the combustion-turbine plant this hour (MW), each between 0 and its plant's capacity.

(a) Write the objective: maximize the utility's profit this hour. Give its units.

Name: _____

(b) Write the three shared limits (river heat, NOx, transmission line) as constraints on y_1, y_2 and y_3. Give the units of each.

Problem 2: Solve It in JuMP

(a) Record the plan `JuMP` finds.

Plant	**Output (MW)**	**At its capacity? (yes/no)**
Coal steam		
Combined cycle		
Combustion turbines		
Profit this hour ($)		

(b) The coal plant is the cheapest to run, yet it is below its capacity, while the combined-cycle plant is at its capacity. Compare the profit each of the two earns per MWth of heat it sends to the river, and use the comparison to explain the plan.

Name: _____

Problem 3: Shadow Prices

(a) Fill in the table.

Constraint	**Amount the plan uses**	**Limit**	**Binding? (yes/no)**	**Shadow price**	**Units of the shadow price**
River heat		1,000 MWth			
NOx		650 lb			
Transmission line		1,000 MW			

(b) Why is the NOx limit's shadow price zero? (One sentence.)

(c) If the transmission line could carry one more MW, which plant would generate it? Use your answer to explain the line's shadow price.

Name: _____

Problem 4: A Hotter River

The next month is warmer, and the river's water temperature is further increased. The state cuts the plants' heat limit from 1,000 to 850 MWth for ecological purposes.

(a) Before you re-solve: use the heat limit's shadow price to predict the utility's profit under the new limit. Show the calculation.

(b) Re-solve in `JuMP` and record the results.

	Heat limit 1,000 MWth	**Heat limit 850 MWth**
Coal steam output (MW)		
Combined cycle output (MW)		
Combustion turbine output (MW)		
Profit this hour ($)		
NOx emitted this hour (lb)		

Does the profit match your prediction from (a)? (One sentence.)

(c) In Problem 2b you found what one more MWth of heat would be worth to the coal plant alone. The heat limit's shadow price is smaller. Using the change in each plant's output from (b), explain why.

(d) The shadow price told you what the cut costs the utility. What does the re-solve show about the cut's effect on the air, and why?